

```

12V1N  protected always-on pull: J1.1, D1, K1.13, K1.A1, D4.K, U2.V1, C5, R1, R14
12V2SW  switched laser after K1.14: C3, J2.1, FAN1-3 pin2, R15
3V3     U2V0 => U1.B, C4, R11, R12, J4.2, R20 => C7
IGN12V  raw 12V ignition after diode-OR (D2/D3 cathodes) ==> R6
IGNSTATE opto output ACTIVE-LOW: U3.A, R12, B13, J3.2
B13/SE1 SE => R2, J2.1, U1.C, U1.D, U1.E, U1.F, U1.G, U1.H, U1.J, U1.K, U1.L, U1.M, U1.N, U1.O, U1.P, U1.Q, U1.R, U1.S, U1.T, U1.U, U1.V, U1.W, U1.X, U1.Y, U1.Z, U1.AA, U1.AB, U1.AC, U1.AD, U1.AE, U1.AF, U1.AG, U1.AH, U1.AI, U1.AJ, U1.AK, U1.AL, U1.AM, U1.AN, U1.AO, U1.AP, U1.AQ, U1.AR, U1.AT, U1.AU, U1.AV, U1.AW, U1.AX, U1.AY, U1.AZ, U1.BA, U1.BB, U1.BC, U1.BD, U1.BE, U1.BF, U1.BG, U1.BH, U1.BI, U1.BJ, U1.BK, U1.BL, U1.BM, U1.BN, U1.BO, U1.BP, U1.BQ, U1.BR, U1.BT, U1.BU, U1.BV, U1.BW, U1.BX, U1.BY, U1.BZ, U1.CA, U1.CB, U1.CC, U1.CD, U1.CE, U1.CF, U1.CG, U1.CH, U1.CI, U1.CJ, U1.CK, U1.CL, U1.CM, U1.CN, U1.CO, U1.CP, U1.CQ, U1.CR, U1.CT, U1.CU, U1.CV, U1.CW, U1.CX, U1.CY, U1.CZ, U1.DA, U1.DB, U1.DC, U1.DD, U1.DE, U1.DF, U1.DG, U1.DH, U1.DI, U1.DJ, U1.DK, U1.DL, U1.DM, U1.DN, U1.DO, U1.DP, U1.DQ, U1.DR, U1.DT, U1.DU, U1.DV, U1.DW, U1.DX, U1.DY, U1.DZ, U1.EA, U1.EB, U1.EC, U1.ED, U1.EE, U1.EF, U1.EG, U1.EH, U1.EI, U1.EJ, U1.EK, U1.EL, U1.EM, U1.EN, U1.EO, U1.EP, U1.EQ, U1.ER, U1.ET, 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U1.KM, U1.KN, U1.KO, U1.KP, U1.KQ, U1.KR, U1.KT, U1.KU, U1.KV, U1.KW, U1.KX, U1.KY, U1.KZ, U1.LA, U1.LB, U1.LC, U1.LD, U1.LE, U1.LF, U1.LG, U1.LH, U1.LI, U1.LJ, U1.LK, U1.LL, U1.LM, U1.LN, U1.LO, U1.LP, U1.LQ, U1.LR, U1.LT, U1.LU, U1.LV, U1.LW, U1.LX, U1.LY, U1.LZ, U1.MA, U1.MB, U1.MC, U1.MD, U1.ME, U1.MF, U1.MG, U1.MH, U1.MI, U1.MJ, U1.MK, U1.ML, U1.MM, U1.MN, U1.MO, U1.MP, U1.MQ, U1.MR, U1.MT, U1.MU, U1.MV, U1.MW, U1.MX, U1.MY, U1.MZ, U1.NA, U1.NB, U1.NC, U1.ND, U1.NE, U1.NF, U1.NG, U1.NH, U1.NI, U1.NJ, U1.NK, U1.NL, U1.NM, U1.NN, U1.NO, U1.NP, U1.NQ, U1.NR, U1.NT, U1.NU, U1.NV, U1.NW, U1.NX, U1.NY, U1.NZ, U1.OA, U1.OB, U1.OC, U1.OD, U1.OE, U1.OF, U1.OG, U1.OH, U1.OI, U1.OJ, U1.OK, U1.OL, U1.OM, U1.ON, U1.OO, U1.OP, U1.OQ, U1.OR, U1.OT, U1.OU, U1.OV, U1.OW, U1.OX, U1.OY, U1.OZ, U1.PA, U1.PB, U1.PC, U1.PD, U1.PE, U1.PF, U1.PG, U1.PH, U1.PI, U1.PJ, U1.PK, U1.PL, U1.PM, U1.PN, U1.PO, U1.PP, U1.PQ, U1.PR, U1.PT, U1.PU, U1.PV, U1.PW, U1.PX, U1.PY, U1.PZ, U1.QA, U1.QB, U1.QC, U1.QD, U1.QE, 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U1.VZ, U1.WA, U1.WB, U1.WC, U1.WD, U1.WE, U1.WF, U1.WG, U1.WH, U1.WI, U1.WJ, U1.WK, U1.WL, U1.WM, U1.WN, U1.WO, U1.WP, U1.WQ, U1.WR, U1.WT, U1.WU, U1.WV, U1.WX, U1.WY, U1.WZ, U1.XA, U1.XB, U1.XC, U1.XD, U1.XE, U1.XF, U1.XG, U1.XH, U1.XI, U1.XJ, U1.XK, U1.XL, U1.XM, U1.XN, U1.XO, U1.XP, U1.XQ, U1.XR, U1.XT, U1.XU, U1.XV, U1.XW, U1.XX, U1.XY, U1.XZ, U1.YA, U1.YB, U1.YC, U1.YD, U1.YE, U1.YF, U1.YG, U1.YH, U1.YI, U1.YJ, U1.YK, U1.YL, U1.YM, U1.YN, U1.YO, U1.YP, U1.YQ, U1.YR, U1.YT, U1.YU, U1.YV, U1.YW, U1.YX, U1.YZ, U1.ZA, U1.ZB, U1.ZC, U1.ZD, U1.ZE, U1.ZF, U1.ZG, U1.ZH, U1.ZI, U1.ZJ, U1.ZK, U1.ZL, U1.ZM, U1.ZN, U1.ZO, U1.ZP, U1.ZQ, U1.ZR, U1.ZT, U1.ZU, U1.ZV, U1.ZW, U1.ZX, U1.ZY, U1.ZZ, U1.AA, U1.AB, U1.AC, U1.AD, U1.AE, U1.AF, U1.AG, U1.AH, U1.AI, U1.AJ, U1.AK, U1.AL, U1.AM, U1.AN, U1.AO, U1.AP, U1.AQ, U1.AR, U1.AT, U1.AU, U1.AV, U1.AW, U1.AX, U1.AY, U1.AZ, U1.BA, U1.BB, U1.BC, U1.BD, U1.BE, U1.BF, U1.BG, U1.BH, U1.BI, U1.BJ, U1.BK, U1.BL, U1.BM, U1.BN, U1.BO, U1.BP, U1.BQ, U1.BR, U1.BT, U1.BU, U1.BV, U1.BW, U1.BX, U1.BY, U1.BZ, U1.CA, U1.CB, U1.CC, U1.CD, U1.CE, U1.CF, U1.CG, U1.CH, U1.CI, U1.CJ, U1.CK, U1.CL, U1.CM, U1.CN, U1.CO, U1.CP, U1.CQ, U1.CR, U1.CT, U1.CU, U1.CV, U1.CW, U1.CX, U1.CY, U1.CZ, U1.DA, U1.DB, U1.DC, U1.DD, U1.DE, U1.DF, U1.DG, U1.DH, U1.DI, U1.DJ, U1.DK, U1.DL, U1.DM, U1.DN, U1.DO, U1.DP, U1.DQ, U1.DR, U1.DT, U1.DU, U1.DV, U1.DW, U1.DX, U1.DY, U1.DZ, U1.EA, U1.EB, U1.EC, U1.ED, U1.EE, U1.EF, U1.EG, U1.EH, U1.EI, U1.EJ, U1.EK, U1.EL, U1.EM, U1.EN, U1.EO, U1.EP, U1.EQ, U1.ER, U1.ET, U1.EU, U1.EV, U1.EW, U1.EX, U1.EY, U1.EZ, U1.FA, U1.FB, U1.FC, U1.FD
```

[illegible]

The schematic diagram illustrates the Ignition Control Module (ICM) circuit for the ATTiny85-20P microcontroller. The circuit is divided into several functional blocks:

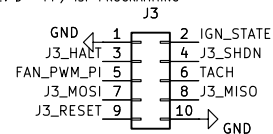
- Input Section:**
 - ACC_IN:** Connected to diode D2 (1N5822) and IGN_12V.
 - ON_IN:** Connected to diode D3 (1N5822) and IGN_12V.
 - 12V_IN:** Connected to capacitor C5 (100nF) to GND and capacitor C4 (100nF) to 3V3.
- Regulation and Power Section:**
 - 3V3:** Provided by a voltage regulator U2 (LM2936Z-3.3) connected to 12V_IN and GND.
 - Capacitors:** C7 (47uF, 25V) is connected to 3V3 and GND. R20 (1R) is connected to 3V3 and C7.
- Microcontroller (U1) Connections:**
 - VCC:** 3V3
 - GND:** GND
 - AREF/PB0:** 5
 - PB1:** 6
 - PB2:** 7
 - XTAL1/PB3:** 2
 - XTAL2/PB4:** 3
 - RESET/PB5:** 1
- Output Section:**
 - COIL_DRV:** Connected to R16 (1k) and LED3 (3mm) to GND.
 - COIL_LOW:** Connected to transistor Q2 (BC337-40) and GND.
 - BATT_SENSE:** Connected to Z2 (BZX55C3V6) and GND.
 - LED3_DRV:** Connected to R11 (10k) and nRESET.
 - COIL_DRV:** Connected to R3 (1k) and Q2_BASE.
 - COIL_LOW:** Connected to R2 (15k) and GND.
 - BATT_SENSE:** Connected to R1 (68k) and 12V_IN.
 - COIL_LOW:** Connected to Q2 (BC337-40) and GND.
 - BATT_SENSE:** Connected to C6 (100nF) and GND.

LED1 update: 12V_IN -> SW1 -> LED_TEST -> R14 -> LED1 -> GND. Hold TEST to check input power.

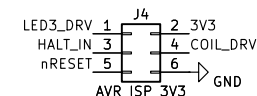
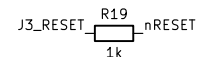
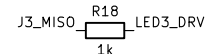
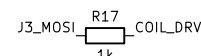
H1 H2 H3 H4

MountingHole MountingHole MountingHole MountingHole

REV D - PI / ISP PROGRAMMING



Pi interface / ISP



J4: 3.3V signals only; target-power OFF.

Pin 2 senses VCC. Fit JP1 to program.

Relay pin map: KiCad A1/A2 = Omron coil 1/5; KiCad 13/14 = Omron contacts 3/4
Nets: 12V_IN = protected always-on rail; 12V_SW = after relay K1; IGN_STATE is ACTIVE-LOW
S660-PDB

Sheet: /
File: S660-PDB.kicad_sch

Title: S660 CarPlay – Power Distribution Board

Size: A4	Date: 2026-09-02
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KiCad E.D.A. 10.0.6

Rev: C

Id: 1/1